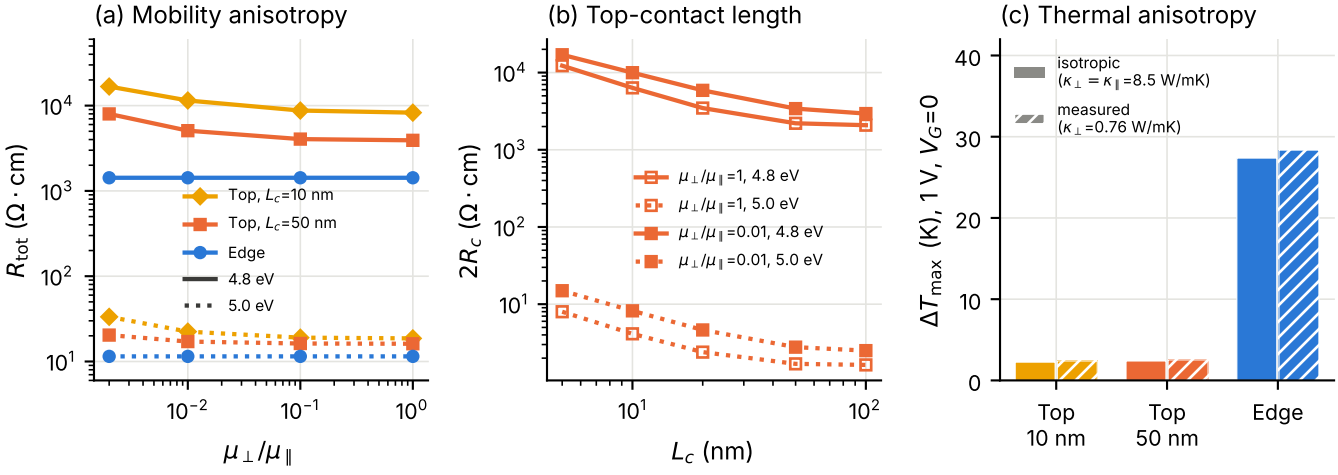
(a) Mobility anisotropy
R_{tot} ($\Omega \cdot$ cm)
10^4
10^3
10^2
10^1
Top, L_c=10 nm
Top, L_c=50 nm
Edge
4.8 eV
5.0 eV
10^{-2}
10^{-1}
10^0
$\mu_\perp/\mu_\parallel$
(b) Top-contact length
$2R_c$ ($\Omega \cdot$ cm)
10^4
10^3
10^2
10^1
$\mu_\perp/\mu_\parallel$=1, 4.8 eV
$\mu_\perp/\mu_\parallel$=1, 5.0 eV
$\mu_\perp/\mu_\parallel$=0.01, 4.8 eV
$\mu_\perp/\mu_\parallel$=0.01, 5.0 eV
10^1
10^2
L_c (nm)
(c) Thermal anisotropy
ΔT_{max} (K), 1 V, V_G=0
40
30
20
10
0
isotropic
($\kappa_\perp = \kappa_\parallel$=8.5 W/mK)
measured
($\kappa_\perp$=0.76 W/mK)
Top
10 nm
Top
50 nm
Edge